

DURHAM 2 SSW, NH, USA

WMO: 726100

Lat: 43.1092N

Lon: 70.9484W

Elev: 19

StdP: 101.09

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 54795

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.5	-24.5	0.4	-15.3	-22.2	0.5	-12.8	5.6	-6.7	5.1	-7.1	1.0	N/A	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	31.3	22.9	29.7	21.7	28.0	20.9	24.1	29.0	23.3	27.9	22.4	26.5	1.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.3	26.8	21.8	16.5	25.6	21.0	15.7	24.6	72.4	29.1	69.0	28.1	65.8	26.3	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	4.0	3.5		-22.8	34.1	3.1	1.2	-25.1	35.0	-26.9	35.7	-28.7	36.3	-30.9	37.2	
			DB	-23.4	25.8	2.3	0.7	-25.0	26.3	-26.3	26.7	-27.6	27.1	-29.2	27.6	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-4.6	-3.4	0.9	7.3	12.8	17.5	21.1	20.2	16.2	9.9	4.0	-1.4
	DBStd	9.92	5.90	5.39	5.12	4.62	4.18	3.74	2.86	3.04	3.89	4.25	4.55	4.92
	HDD10.0	1831	452	375	284	107	17	1	0	0	1	56	186	353
	HDD18.3	3839	710	608	539	331	180	59	7	15	87	263	429	611
	CDD10.0	1264	1	0	3	27	103	224	344	316	187	51	7	1
	CDD18.3	233	0	0	0	1	8	34	93	73	23	1	0	0
	CDH23.3	2255	0	0	4	41	139	354	855	641	208	12	0	0
	CDH26.7	646	0	0	1	12	39	100	261	179	53	1	0	0
(14) Wind	WSAvg	1.2	1.4	1.5	1.7	1.6	1.3	1.0	0.9	0.9	0.9	1.1	1.2	1.3
(15) Precipitation	PrecAvg	1032	77	74	79	94	85	81	77	78	84	98	107	97
	PrecMax	1494	199	262	233	291	237	286	200	224	236	354	286	237
	PrecMin	626	12	0	24	14	18	21	15	21	7	5	16	12
	PrecStd	187	44	41	37	53	45	51	42	41	46	70	54	53
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	14.9	21.4	28.4	30.6	32.3	33.0	33.0	31.7	25.0	20.0	14.5
		MCWB	12.7	10.0	13.7	17.3	20.1	22.4	24.1	24.6	23.4	19.1	15.4	12.4
	2%	DB	9.2	10.2	15.1	22.9	27.2	29.6	31.1	30.5	28.1	22.1	16.9	10.6
		MCWB	6.5	6.1	9.9	15.4	18.4	21.4	23.1	23.1	21.6	17.5	12.8	8.6
	5%	DB	5.9	6.7	11.4	19.3	24.2	27.2	29.6	28.6	25.6	19.4	14.2	8.3
		MCWB	3.2	3.5	6.4	13.5	16.9	19.8	22.0	21.4	20.1	16.2	10.0	6.4
	10%	DB	3.4	4.1	8.3	16.2	21.3	25.2	27.9	26.9	23.5	17.0	11.8	5.7
		MCWB	1.2	1.3	4.7	9.9	15.9	18.4	21.4	20.4	19.3	14.1	8.8	3.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	12.9	15.0	18.2	21.8	23.9	25.6	25.0	24.3	20.9	16.5	12.5
		MCDB	14.2	15.2	22.7	22.2	28.8	29.3	31.3	30.7	29.0	23.2	19.4	13.2
	2%	WB	7.3	7.8	11.8	16.2	20.2	22.7	24.5	23.7	22.8	18.5	13.5	9.4
		MCDB	8.4	10.9	15.6	21.8	25.5	28.1	29.8	28.1	26.9	20.3	15.8	10.3
	5%	WB	3.9	4.5	7.4	13.6	18.5	21.4	23.6	22.6	21.7	16.8	11.0	7.2
		MCDB	5.5	7.6	10.4	17.9	23.0	25.5	28.3	26.5	25.1	19.2	13.5	8.5
	10%	WB	1.7	2.6	4.9	10.7	16.6	19.9	22.7	21.6	20.4	15.0	8.9	4.5
		MCDB	3.4	4.7	7.8	14.7	20.9	23.9	26.9	24.9	22.9	17.2	11.1	6.1
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.2	10.3	11.7	11.5	11.5	11.6	12.0	12.1	10.9	10.0	8.8
		MCDBR	11.6	13.7	14.7	17.9	16.5	14.6	13.9	14.0	14.5	13.7	13.9	11.8
	5% WB	MCWBR	10.5	9.8	9.6	9.7	8.4	7.4	6.7	6.8	7.8	8.6	10.4	9.5
		MCWBR	11.5	13.6	14.3	14.1	14.5	12.9	12.0	11.4	11.8	11.3	12.6	10.7
(40) Clear-Sky Solar Irradiance	taub	taub	0.286	0.297	0.321	0.358	0.395	0.441	0.461	0.428	0.374	0.334	0.312	0.292
		taud	2.470	2.454	2.471	2.415	2.350	2.251	2.204	2.284	2.416	2.530	2.550	2.538
	Ebn at Noon	Ebn at Noon	859	908	925	909	879	836	813	830	854	854	824	823
		Edh at Noon	72	87	97	111	122	135	140	125	102	80	67	63
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.71	2.58	3.64	4.61	5.13	5.52	5.81	5.27	4.24	2.75	1.84	1.38
	RadStd	RadStd	0.11	0.20	0.23	0.50	0.59	0.49	0.38	0.26	0.26	0.25	0.16	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+33

Nomenclature: See separate page